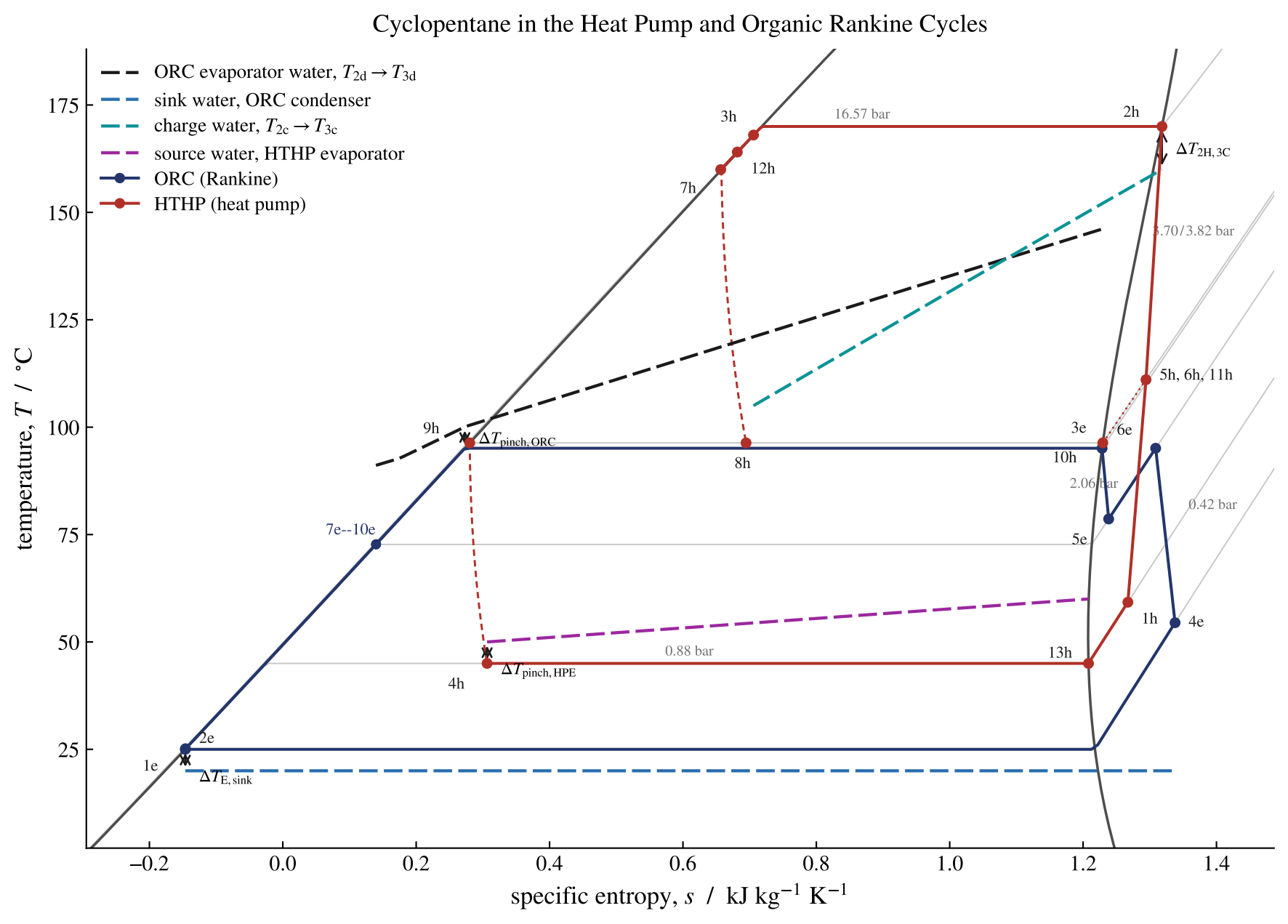
Cyclopentane in the Heat Pump and Organic Rankine Cycles
ORC evaporator water, $T_{2d} \rightarrow T_{3d}$
sink water, ORC condenser
charge water, $T_{2c} \rightarrow T_{3c}$
source water, HTHP evaporator
ORC (Rankine)
HTHP (heat pump)
temperature, T / °C
specific entropy, s / kJ kg^{-1} K^{-1}
175
150
125
100
75
50
25
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
16.57 bar
3.70/3.82 bar
2.06 bar
0.88 bar
0.42 bar
$\Delta T_{2H,3C}$
$\Delta T_{pinch,ORC}$
$\Delta T_{pinch,HPE}$
$\Delta T_{E,sink}$
1e
2e
2h
3h
4e
4h
5e
5h, 6h, 11h
6e
7e--10e
7h
8h
9h
10h
12h
13h
1h
3e